

voltage modes. Separate three-digit displays for voltage and current, with individual control of voltage and current, are some interesting features in 1415.

Programmable supply with USB connectivity. May be due to the Internet of Things (IoT) coming into the picture, or some other reason, everybody wants to control these devices from their computers. The programmable feature allows remote control of their operation through an analogue input or digital interface. "The latest versions have various interfaces like LAN and RS232," says Gupta. Applications include automated equipment testing, crystal growth monitoring, semiconductor fabrication and X-ray generation. 1350 DC power supply from Global Specialties features programmable control via USB interface and microprocessor-controlled circuits.

National Instruments PXI-4110 is a programmable, triple-output precision DC power supply in a single slot with three analogue channels in PXI module. It has 16-bit resolution for programming the voltage set point and current limit.

In case of the Z+ series from TDK Lambda, connectivity options include built-in USB, RS232 and RS485 interfaces, along with LAN, GPIB and isolated analogue interfaces. Z+ is light as well; you can move the power supply around without hitting the gym.

Not a feather but still light.

Benchtop equipment are generally heavier than other types, which re-

Trends at a glance

- Newer supplies offer resolution as low as 1mA and 1mV
- Simulate a battery so that performance can be verified
- Adjustable internal resistance, slew rate and inrush current
- Multiple interfaces like LAN and RS232
- Analogue control makes these useful for automated test setups as well
- Higher-end supplies can source as well as sink current

quires moving the test setup to the supply every time. However, Gupta says, "The devices are becoming smaller and more energy-efficient." REOLINE range of portable power supplies provides an adjustable low-loss output voltage, in addition to being lightweight. The lightweight equipment are certainly more portable. These might not be as portable as handheld equipment but the move has certainly begun.

A case in point would be the benchtop PSU 130 from Jameco Electronics. This single-channel supply is a compact little device weighing about 500gm. With two digital four-digit displays, it measures less than 15.2cm (6-inch). It also features short-circuit and overload protection.

Safety in equipment. One of the sure-fire ways to damage a power supply is to overload it. Excess current heats up the components, which eventually fail. Components ranging from transistors and diodes to resistors and capacitors could easily burn out on overload. This calls for some safety features to be built into the device.

Among the various power supplies in power supplies released recently would be REOLINE range of portable power supplies from REO UK Ltd. It features step-less adjustment of voltage from absolute zero to maximum output. Safety features include operation under

high inrush currents as well as ability to withstand sustained overloads, without a detrimental impact on overall performance. "Research labs and test departments simply cannot operate effectively without variable voltage," says Steve Hughes, managing director, REO UK.

Safety features in Genesys series from TDK Lambda include a safe start, which returns the power supply to its last setting configuration post power interruption. Other features include over-voltage protection, under-voltage limit, fold-back protection and over-temperature protection.

Supplying the required power.

"Power supplies are going to stay as tools for product design, testing, QA, production as well as service applications," says Gupta. If recent advances are any indication, it seems all uphill from here.

Rao says, "In the future, users will require multiple-channel versions of these supplies to address banks of batteries." He adds, "Battery simulators will incorporate more sophisticated models to simulate battery performance."

With computer connectivity through LAN and RS232, these power supplies have made operation easier as these save time and hassle. According to Siglent, USB, LAN and other interfaces will become standard functions in power supplies. The circuit design of power supplies should be optimised to achieve better dynamic performance.

If innovations keep happening at this pace, we will probably have some more interesting features in the near future. **EFY**



Fig. 2: A system sourcemeter